

BC 7. For what value of k will $x + \frac{k}{x}$ have a relative maximum at $x = -2$?

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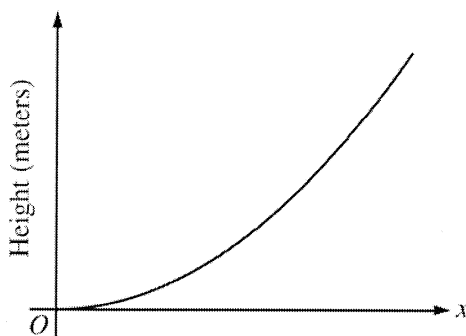
- (A) -4 (B) -2 (C) 2 (D) 4 (E) None of these

BC 18. Let g be a continuous function on the closed interval $[0,1]$. Let $g(0) = 1$ and $g(1) = 0$. Which of the following is NOT necessarily true?

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- (A) There exists a number h in $[0,1]$ such that $g(h) \geq g(x)$ for all x in $[0,1]$.
 (B) For all a and b in $[0,1]$, if $a = b$, then $g(a) = g(b)$.
 (C) There exists a number h in $[0,1]$ such that $g(h) = \frac{1}{2}$.
 (D) There exists a number h in $[0,1]$ such that $g(h) = \frac{3}{2}$.
 (E) For all h in the open interval $(0,1)$, $\lim_{x \rightarrow h} g(x) = g(h)$.



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3. The figure above is the graph of a function of x , which models the height of a skateboard ramp. The function meets the following requirements.

- (i) At $x = 0$, the value of the function is 0, and the slope of the graph of the function is 0.
 (ii) At $x = 4$, the value of the function is 1, and the slope of the graph of the function is 1.
 (iii) Between $x = 0$ and $x = 4$, the function is increasing.
- (a) Let $f(x) = ax^2$, where a is a nonzero constant. Show that it is not possible to find a value for a so that f meets requirement (ii) above.
- (b) Let $g(x) = cx^3 - \frac{x^2}{16}$, where c is a nonzero constant. Find the value of c so that g meets requirement (ii) above. Show the work that leads to your answer.
- (c) Using the function g and your value of c from part (b), show that g does not meet requirement (iii) above.
- (d) Let $h(x) = \frac{x^n}{k}$, where k is a nonzero constant and n is a positive integer. Find the values of k and n so that h meets requirement (ii) above. Show that h also meets requirements (i) and (iii) above.